



SSIS ETL Project – Sales Data

1. Overview

You are required to build a complete ETL project using SSIS that loads sales data from CSV files into a database on SQL Server, while also making use of the Products, Customers, Categories, and Branches data from the previous session.

Complete the assignment based on the concepts covered in the session. You're free to choose your own implementation approach, unless a specific method is explicitly stated below.

2. Source Data

- **Sales data (Orders):** CSV files — Orders_May, Orders_June, Orders_July — these are the core data for this task.
- **Reference data from the previous session:** Products, Customers, Categories, Branches — used for lookups/joins with the sales data where needed.
- **Destination:** a new database created on SQL Server, with the data loaded into an appropriate table/tables.

3. Requirements

A. Dynamic File Handling

- The project must be able to read all sales files in a single folder without modifying the package when new files are added.
- Use Variables to control the values used during execution (e.g. the current file path).
- Use Expressions to link these variables to the properties of the Flat File Source's Connection Manager (Dynamic Flat File Source), so the file path read at runtime changes automatically.

B. Processing Multiple Files

- Use a ForEach Loop Container to iterate over all CSV files in the folder, ensuring each file is processed exactly once.
- The project must be able to handle any new files added to the same folder later, without any changes to the package.



C. File Management

- After each file is processed successfully, build a step in the pipeline that moves the file to another folder (Archive/Processed) or renames it to indicate it has been processed.
- This step should rely on the same Variables and Expressions used to read the file, so it works dynamically for whichever file is running.

D. Using a For Loop Container

- Implement part of the project using a For Loop Container, using Variables to control the number of iterations or the values used within each cycle.

E. Data Quality

- Validate the data before loading it into the database.
- Any invalid data must be handled appropriately and kept for later review (not discarded).

F. Data Profiling

- Run a Data Profiling process on the data before loading, and produce a report showing data quality and some characteristics of the columns used.

G. Data Processing / Summarization

- Apply transformations to the sales data (e.g. summarization/aggregation) and join it with Products, Customers, Categories, and Branches data where needed.

H. Notification

- After the package runs successfully, send a notification email with a summary of the execution. You may use either:
- Send Mail Task, or
- A Script Component with C# code that sends the email directly (e.g. Gmail-to-Gmail), since Send Mail Task only works between mail servers.
- If execution fails, send a notification (via either method above) indicating an error occurred while running the package.



4. Bonus Task — Working with APIs

This part is optional and counts as extra credit (Bonus), based on the attached APIs.txt file.

- Choose any API from the file (or another suitable API) and use it.
- Build a full pipeline that clearly shows the Source, the Transformations in between, and the Destination.
- To connect to the API, use a Script Component and write C# code inside it to make the API calls.

5. Deliverables

1. Complete SSIS project.
2. The destination database.
3. All CSV files used.
4. The resulting Data Profiling report.
5. Screenshots showing successful execution.
6. A short write-up (max one page) covering: how Variables were used, how Expressions were used, the mechanism for processing files in the folder, the role of the Foreach Loop Container vs. the For Loop Container, and the file management mechanism after loading.

6. Final Validation Checklist

- All files in the Source folder were loaded.
- No file was loaded more than once.
- All data was successfully loaded into the database.
- Processed files were handled per the implemented file management mechanism.
- The Data Profiling report was successfully generated.
- End-of-execution notifications (success/failure) were sent successfully.